

Curriculum Vitae

Jessica O. Diallo

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Education

2020 – Present. PhD in Aquatic and Fishery Sciences,

University of Washington. Freshwater Ecology and Conservation Lab with Dr. Julian Olden.

Data Science Option. Future Rivers Program. Cumulative GPA: 3.96/4.0.

2018 – 2019. Coursework in Math and Statistics (non-matriculated),

University of Washington. Probability I and II, Matrix Algebra, Advanced Multivariate Calculus, Differential Equations.

2013. B.S. in Biology and Environmental Studies (double major) with Comprehensive Honors, University of Wisconsin-Madison. Cumulative GPA: 3.9/4.0.

Research Experience

2020 – Present. PhD candidate. University of Washington School of Aquatic and Fishery Sciences. Researching the food web impacts of nonnative fishes in the Bill Williams River and evaluating the success of an invasive fish removal program in Julian Olden's lab.

2023 – 2024. Graduate Student Intern. NOAA Northwest Fisheries Science Center. Working with Beth Sanderson and other NOAA scientists to evaluate the Northern Pikeminnow Management Program.

2016. Environmental Intern. King County Department of Natural Resources and Parks. Sampled macroinvertebrate communities in urban and rural streams throughout King County, contributing to Pacific salmon research and habitat restoration efforts. Conducted aquatic and terrestrial habitat surveys, interacted with public citizens while in the field, and organized data for Puget Sound Stream Benthos database.

2013. Research Intern. Smithsonian Environmental Research Center. Studied the effects of invasive marsh grass habitat alterations on predator-prey interactions in Chesapeake Bay estuaries, responsible for a high to low tide relative predation experiment. Caught and identified marine fishes and invertebrates, analyzed and presented research to the public.

2012 – 2013. Senior Honors Thesis. UW-Madison. Behavior and development of caddisfly larvae in response to desiccation stress research continued from summer 2012.

2012. Advanced Independent Research Program. Rocky Mountain Biological Laboratory.

Examined aggressive behavior, cannibalism, and development of caddisfly larvae in response to simulated drying ponds, mentored by Scott Wissinger and Bobbi Peckarsky.

2011 – 2012. Laboratory Technician. UW-Madison. Dissected juvenile Northern Pike for directed study in Peter McIntyre's lab, contributing to research on their spawning migrations in Lake Michigan tributaries of Green Bay, WI.

2010 – 2011. Undergraduate Research Scholars Program. UW-Madison. Studied *Salmonella enterica* genes required for survival on lettuce in Jeri Barak's plant pathology lab.

2010. Research Experience for Undergraduates. Portland State University. Renewable petroleum project involving biochemical examination of the catalytic role of microbe-derived proteins, funded by the National Science Foundation.

Awards and Grants

- 2021. Conservation Research Grant. North American Native Fishes Association (\$1000).
- 2021. Conservation Grant. Desert Fishes Council (\$1000).
- 2021. Small Project Grant. American Fisheries Society – Western Division (\$1000).
- 2020. Award. NSF Graduate Research Fellowship Program.
- 2019. Honorable mention. NSF Graduate Research Fellowship Program.
- 2013. Library Research Award (\$500). UW-Madison.
- 2012. Mark Mensink Honors Research Grant Award (\$7500). UW-Madison.
- 2011. Honors Summer Sophomore Research Apprenticeship Grant (\$2000). UW-Madison.
- 2011. John H. Nelson Undergraduate Research Award (\$500). UW-Madison.

Publications

Published:

Lund, J.O., Wissinger, S.A. & Peckarsky, B.L. 2016. Caddisfly Behavioral Responses to Drying Cues in Temporary Ponds: Implications for Effects of Climate Change. *Freshwater Science* 35(2): 619-630. [maiden name is Lund]

In Press:

Diallo, J.O., Converse, S.J., Chmiel, M., Stites, A., & Olden, J.D. Optimizing control of a freshwater invader in time and space. *Ecological Applications*.

Diallo, J.O. & Olden, J.D. Invasive species invoke a lifetime of trophic change in native stream fishes. *Ecosphere*.

Presentations

2025. “Lifetime Trophic Ecology of Fishes and Invasive Species Management in Dryland Streams.” UW School of Aquatic and Fishery Sciences PhD Defense.

2024. “Predation and Predator Management.” NOAA Pacific Salmon Lifecycle Modeling Workshop #2 in Seattle, WA.

2024. “Not all those who wander are lost: leveraging PIT tag data to better understand northern pikeminnow movement in the Columbia River Basin.” WA/BC & ID Chapters – American Fisheries Society Annual Conference.

2023. “Optimizing invasive species eradication in time and space: a case study of green sunfish removal in intermittent streams.” Western Division – American Fisheries Society Annual Conference.

2023. “Optimizing invasive species eradication in time and space: a case study of green sunfish removal in intermittent streams.” University of Washington School of Aquatic and Fishery Sciences Quantitative Seminar.

2022. “Fish invaders cause a lifetime of trophic change in native desert fishes.” University of Washington School of Aquatic and Fishery Sciences Graduate Student Symposium.

2022. “Fish invaders cause a lifetime of trophic change in native desert fishes.” Joint Aquatic Sciences Meeting hosted by the Society for Freshwater Sciences. Virtual presentation.

2013. “Relative predation risk between native and invasive (*Phragmites*) Marshes in the Chesapeake Bay.” Smithsonian Environmental Research Center Fall Intern Seminar.

2013. “Caddisfly behavioral responses to drying cues in temporary ponds.” UW-Madison Undergraduate Research Symposium. UW-Madison L&S Honors Program Senior Honors Thesis Symposium. Wisconsin Ecology Symposium (graduate level).

2012. “Caddisfly behavioral responses to drying cues in temporary ponds.” Rocky Mountain Biological Laboratory Undergraduate Symposium.

2011. “*Salmonella enterica* genes required for survival on lettuce.” Annual UW-Madison Food Research Institute Conference.

2010. “Renewable Petroleum: Determining the catalytic role of Ole A-D protein.” Portland State University Research Experience for Undergraduates Symposium.

Other relevant professional employment or experience

2025. Teaching Assistant. University of Washington School of Aquatic and Fishery Sciences. Freshwater Ecology and Conservation course taught by Julian Olden. Independently taught and coordinated lab section of 17 undergraduate students. Lab material included river and lake hydrology, biomonitoring using macroinvertebrates, and river restoration.

2016 – 2021. Administrative Specialist. Peace Corps Office of Volunteer Recruitment and Selection. Managed purchase and travel funds totaling over \$35,000 per quarter, assisted the Administrative Officer with preparing and executing the budget and managing records, conducted staff onboarding and exiting procedures, and trained as a Certified Contracting Officer’s Representative.

2014 – 2015. Child Nutrition Specialist. Peace Corps Response in Mali, West Africa. Collaborated with local health center to curtail child malnutrition using Moringa tree planting and improved porridge demonstrations. Evacuated due to security concerns.

2013 – 2014. Agroforestry Extension Agent. Peace Corps in Guinea, West Africa. Encouraged sustainable agriculture, responsible forestry, beekeeping and environmental education. Evacuated due to Ebola outbreak.

Leadership and Outreach

2023. Mentor for undergraduate student volunteers conducting stomach contents analysis over several months.

2022. Mentor for NSF Graduate Research Fellowship Program Workshop, hosted by UW SAFS.

2022. Planning committee for UW SAFS Graduate Student Symposium.

2022. Volunteer for the UW School of Aquatic and Fishery Sciences Open House.

2021 – 2022. Peer reviewer for the Food Webs journal.

2021. Mentor for an American Fisheries Society Hutton Scholar. Worked directly with a high school student over several months on lab research.

2017 – 2019. Team leader for several projects within my job at the Peace Corps: SharePoint Administrative Resource Site design and creation, Go Animate training video production for purchasing process, and records management project and shared drive merger.

2014. Presented my Peace Corps-Guinea experience to 10 classrooms in three elementary and high schools, including stories and pictures. I also used my blog, published while serving in Guinea, to connect with students from my old high school and share my experience abroad.

Notable computer skills

Microsoft Suite, Access; R Statistical Program; ArcGIS; SharePoint; GitHub